System Design: Monolithic vs Distributed Systems

Easy lecture notes | Simple English | Java developer interview revision

Lecture idea: A monolithic application runs its modules together as one deployable unit. A distributed system uses multiple independent computers or processes that communicate over a network.

1. Monolithic System - Simple Meaning

Think of one large shop where billing, orders, and inventory are handled in the same building. In software, the features are packaged and deployed together.

Example: A Java Spring Boot application contains Login, Orders, Payments and Notifications in one application. Usually the whole application is deployed together.

2. Distributed System - Simple Meaning

Think of several shops working together by phone. In software, multiple computers or services communicate through a network to complete a task.

Example: Order Service runs on Machine A, Payment Service on Machine B, and Notification Service on Machine C. They exchange messages or API requests.

3. Diagram: How They Differ

MONOLITHIC (one deployable app)	DISTRIBUTED (multiple nodes)
[Login | Orders | Payments]	[Order] <-> [Payment]
One application	<-> [Notification]
One deployment	Network communication

4. Why Use a Distributed System?

Horizontal scaling: Add more machines to handle more traffic. For example, one server handles 1,000 requests; adding another can increase total capacity when work is distributed properly.

Fault tolerance: If one machine fails, another may continue the work if redundancy and failover have been designed.

Resource sharing: Machines and services can communicate and share work or data over a network.

5. What Is a Single Point of Failure (SPOF)?

A single point of failure is one component whose failure stops an important part of the system. A single application instance can be a SPOF. Distributed systems can reduce this risk, but do not automatically remove it: a shared database or network can still fail.

6. Replication - How to Reduce Data Loss

Replication means keeping copies of data on more than one machine. If one machine fails, another copy may still be available. Example: Database A has customer records and Database B holds a replica.

Important: Replication helps availability, but correct failover, synchronization and backups are still needed. It does not guarantee zero data loss.

7. Advantages and Disadvantages of Distributed Systems

Advantages: Can scale by adding machines; can improve availability with redundancy; can divide work among multiple nodes.

Disadvantages: More network communication; harder monitoring and debugging; more complex security, deployment, data consistency and failure handling; potentially higher operational cost.

8. Quick Comparison

Feature	Monolithic	Distributed
Deployment	Together	Multiple nodes/services
Network calls	Often fewer internal calls	More network communication
Scaling	Often scale whole app	Can scale parts independently
Failure handling	One instance may stop all	Can isolate failures if designed
Complexity	Simpler to start	Harder to operate

9. Interview Questions - Short Answers

Q1. What is a distributed system? A collection of independent computers or services connected by a network that work together.

Q2. What is horizontal scaling? Adding more machines or instances to increase capacity.

Q3. What is replication? Maintaining copies of data across machines to improve availability and resilience.

Q4. Does a distributed system have no SPOF? Not necessarily. It must be designed with redundancy and failover.

Q5. Which is better? It depends. Monoliths are often simpler for smaller applications; distributed systems can help when scale and independent operation justify added complexity.

10. Remember in 20 Seconds

Monolithic = one deployable application.
Distributed = multiple connected machines/services.
Horizontal scaling = add more machines.
Replication = keep copies of data.
Fault tolerance = keep working despite some failures.

Source note: Based on the Hindi lecture transcript supplied in chat (approximately 0:00-5:02). Some transcript phrases were garbled; explanations are simplified and technical caveats are added for accuracy.